

Object Detection and Recognition Report



Abstract

This report evaluates computer vision techniques for the unconstrained recognition of handwritten numerals. The study first compares a traditional approach using Hu Moments and k-Nearest Neighbour (k-NN) classification against a deep learning Convolutional Neural Network (CNN). Results demonstrated that while Hu Moments performed adequately in controlled lighting, they failed to generalize to style variations and perspective distortion (~30% accuracy). Conversely, the CNN achieved 63.2% accuracy on a complex dataset, exhibiting superior invariance to local deformations.

An integrated detection pipeline was developed to locate and sequence multi-digit strings using adaptive thresholding and spatial sorting. While effective on clean samples, critical analysis revealed that the system's reliance on geometric heuristics (blob analysis) led to false positives in noisy environments, often misclassifying structural noise (such as paper tears) as digits. The project concludes with a design specification for an autonomous mobile robot, proposing the adoption of semantic object detection (YOLOv8) and RGB-D depth sensing to address these environmental robustness challenges.



Introduction



Handwriting recognition remains a fundamental challenge in computer vision, requiring systems that can adapt to high variability in style, lighting, and perspective. This project investigates the evolution of recognition pipelines, comparing traditional feature extraction against modern deep learning architectures.

The primary aims and objectives of this project are:

- Evaluate the robustness of Hu Moments and k-Nearest Neighbour (k-NN) classifiers under varying environmental conditions.
- Design and Train a Convolutional Neural Network (CNN) to improve classification accuracy on complex datasets.
- Develop an integrated pipeline for the simultaneous localization and sequencing of multi-digit strings in unconstrained environments.
- Propose a deployment specification for an autonomous mobile robot (Kobuki) utilizing these vision systems.

Traditional Machine Learning

Method

To evaluate the effectiveness of using a k-Nearest Neighbour (k-NN) classifier on Hu moments, a dataset of handwritten numerals (0-9) was captured under several different conditions:

1. Variation in handwriting style: Numerals written by multiple subjects using varied instruments (biro, marker, pencil) and handedness.
2. Camera angle: The same set of data taken at a 45-degree angle camera shot.
3. Varied lighting: A control set of the ball-point pen numerals taken under dim and uneven lighting.

The images were pre-processed using adaptive thresholding, with a 0.5 sensitivity, to separate the foreground and backgrounds. This was then followed by a morphological opening (bwareopen) to remove noise. Seven Hu Moments were then extracted from each binary image to form a feature vector invariant to translation, scale and 2D rotation. A k-NN classifier, with $k=3$ Euclidian distance, was trained on 70% of the data and tested on the remaining 30% for each condition.

Results and discussion

The systems performance varied significantly across the different testing conditions, showing some of the main disadvantages to using Hu moments.

Table 1: Sensitivity analysis of Hu moments

Condition	Accuracy	Observation
Varied lighting	61.7%	Best performance. The pre-processing successfully normalised the lighting, allowing the Hu moments to work reasonably well.
Variation in handwriting style	30%	Poor performance and failure. Hu Moments struggled to generalize across different pen thicknesses and handwriting irregularities.
Camera Angle	27.7%	Worse performance. Hu moments are rotation-invariant for 2D but perform worse under 3D skew.

The performance and accuracy were the best for the Variable lighting dataset (Figure 1). This shows that if the handwriting is consistent, Hu moments can identify the digits even in bad lighting.

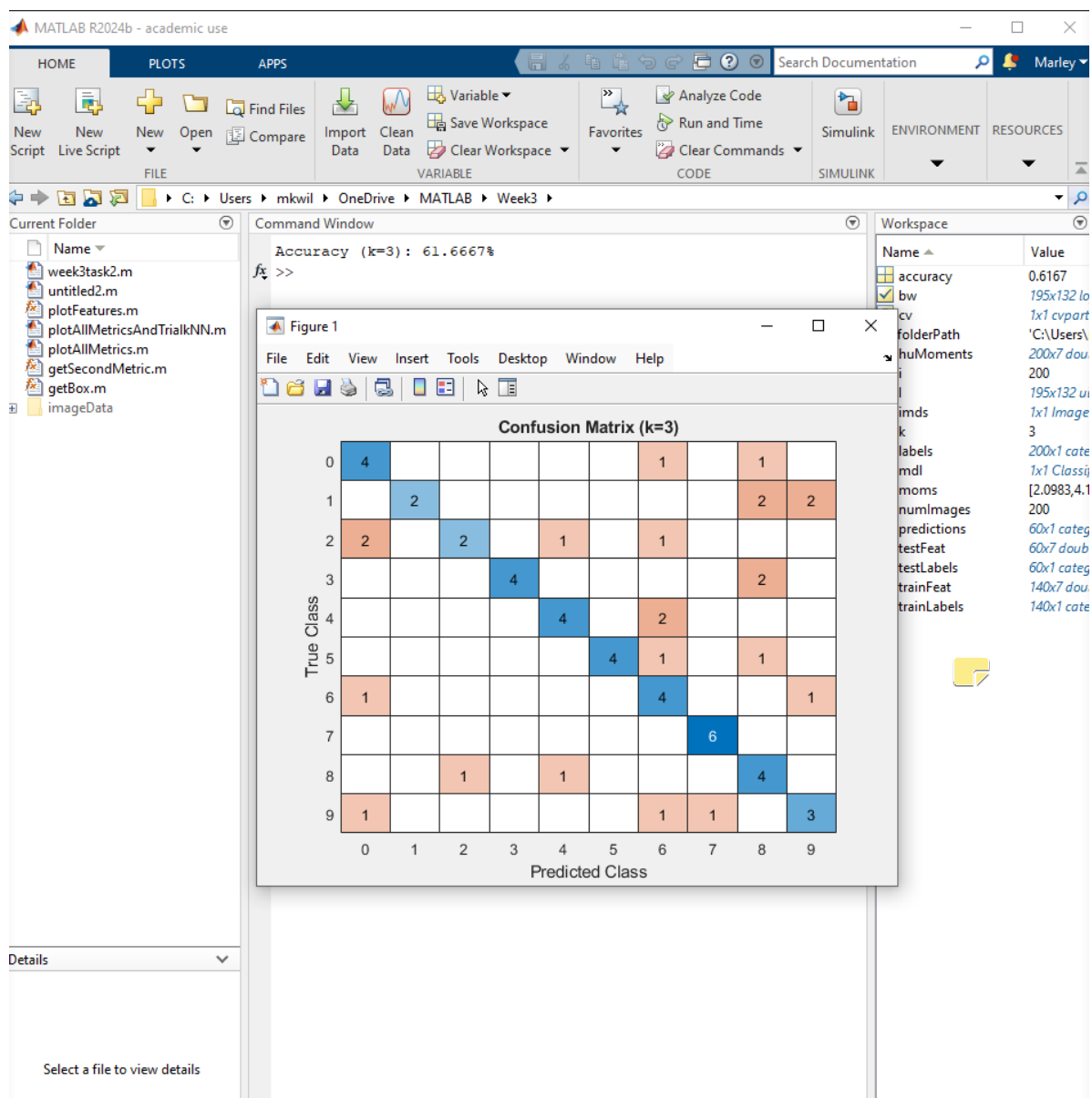


Figure 1

However, this method failed when facing variation in the styles of the numbers written (Figure 2). This is a key finding because the Hu moments rely on calculating the distribution of mass around the centre. The sharpie and pencil/pen have different mass distributions, even if they are both drawn similarly. As well as this the irregularities of different subjects handwriting can drastically change features. This proves that the Hu moments lack ability to recognise numbers written by different people in different handwriting styles.

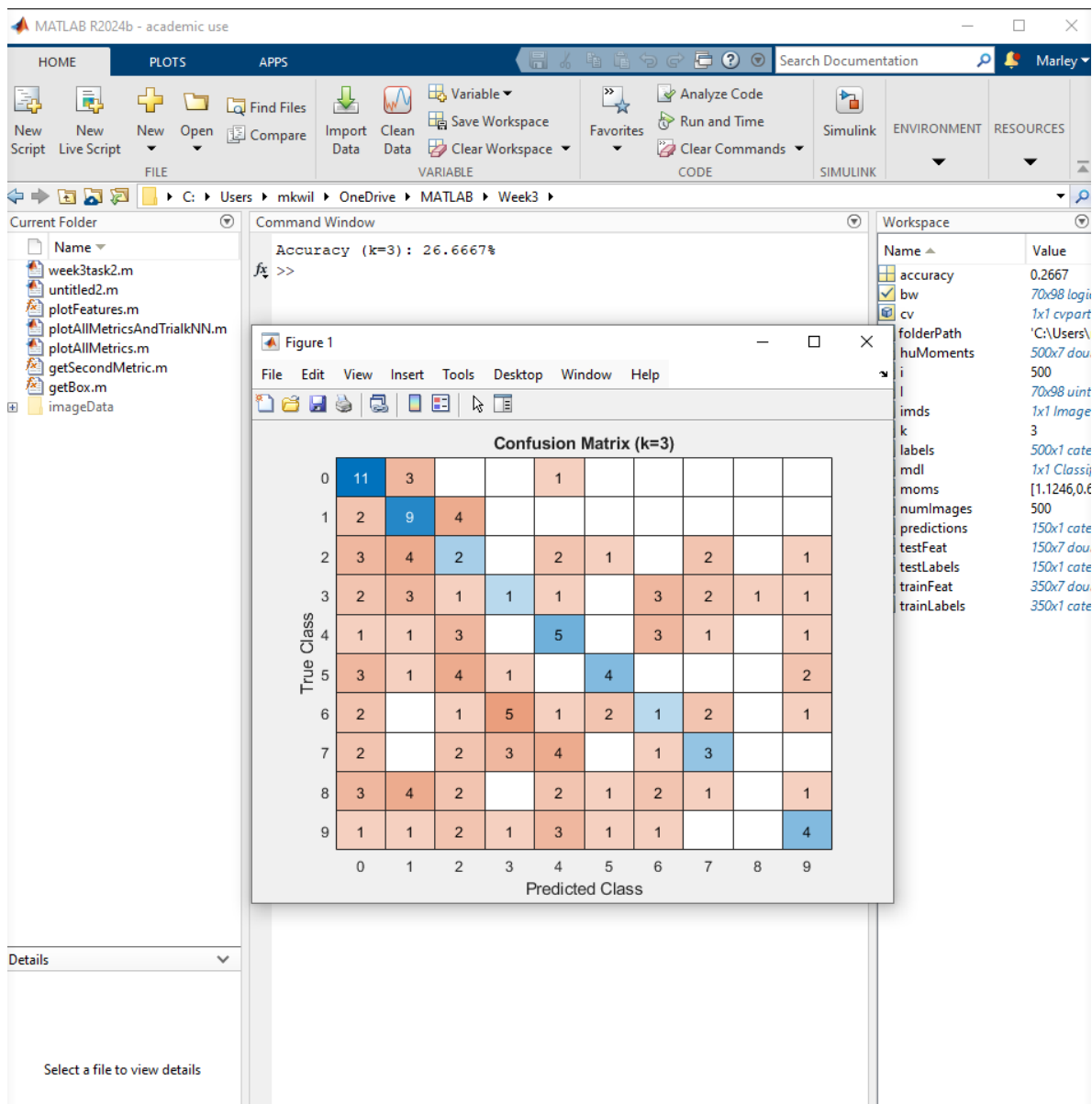


Figure 2

Performance was lowest on the angled dataset (Figure 3). This proves that whilst Hu moments don't change when rotated on a 2D plane, they are not invariant to perspective change and distortion. The 45-degree angle stretches the shape, making the standard moments less effective.

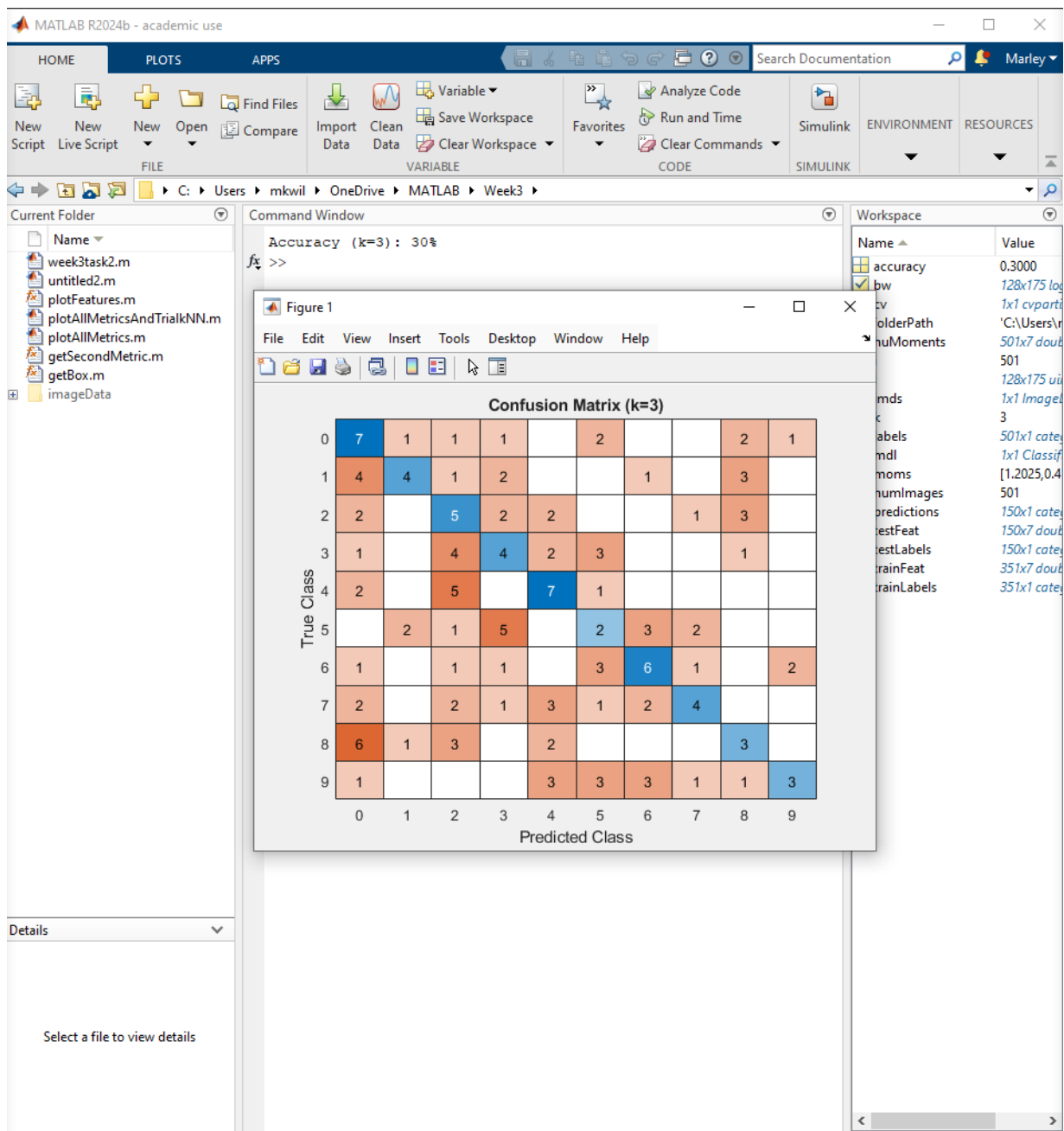


Figure 3

Deep learning for numeral recognition

Method

To evaluate the effectiveness of numeral recognition algorithms, a deep learning approach was implemented using a Convolutional Neural Network (CNN).

To ensure the model was robust enough to different variations, the dataset was constructed by combining all available image sets (standard, angled, rotated, varied writing styles, and lighting) into a single dataset. All of the images were resized to the

standard 28x28x3 pixels to ensure consistency for the input layers. The data was then split into 70% training and 30% testing sets.

The CNN architecture consisted of two convolutional blocks (8 and 16 filters, 3x3) with Batch Normalization, ReLU activation, and Max Pooling (2x2). A fully connected layer mapped extracted features to the 10 output classes via Softmax classification.

The was trained using Stochastic Gradient Descent with Momentum (SGDM) for 10 epochs.

Results

The Deep learning model had a classification accuracy of 63.2%, which was a very large improvement compared to the Hu moment methods used before.

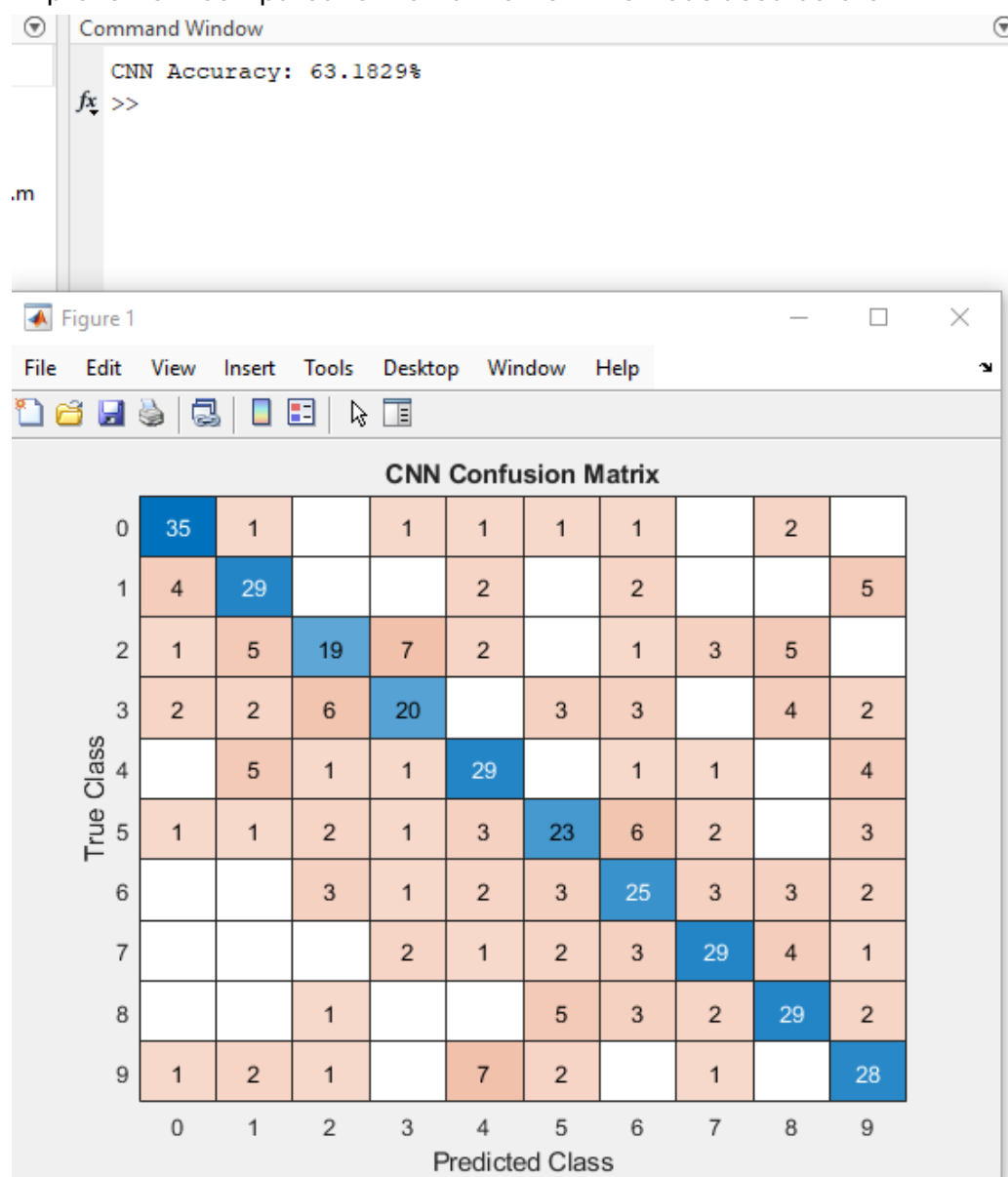


Figure 4

Figure 4 shows the confusion matrix for the CNN. Unlike the Hu moments method, which failed when tested with the varied style datasets, the CNN showed a lot better stability in its model. The matrix shows a much more consistent diagonal, indicating that the network successfully generalised throughout the different variations. Although some confusion still occurred between the digits that are similarly drawn due to the high variance in the dataset.

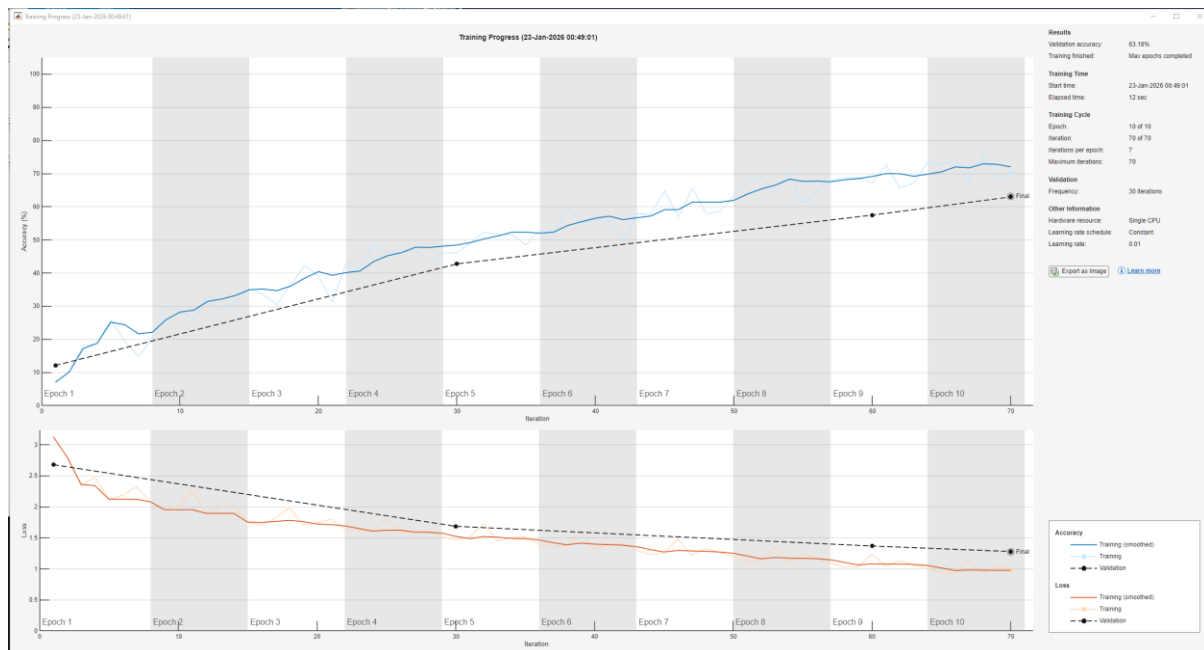


Figure 5

The training process (Figure 5) shows a steady upward trend in validation accuracy. The curve did not curve as it reached epoch 10 suggesting that the model was still learning and that further training epochs may have even higher accuracy.

Comparison to the traditional method

The deep learning approach is much more effective and accurate when comparing the mixed dataset. The CNN outperformed the traditional method by over 30% on the mixed dataset, demonstrating that hierarchical feature learning handles variance better than static shape descriptors. The clear difference between the accuracy of the two shows some of the weaknesses of global shape descriptors which are: they don't work well when facing 3d angles or different widths from the pens. The CNNs ability to learn hierarchal local features allowed it to ignore the style variations and focus on the structure of the digits. Although it is important to note that the Hu moments were capable of good performance when tested on a more consistent dataset. Proving that Hu moments do perform well enough for simpler tasks and more restricted tasks.

Simultaneous detection and classification

Method

The detection pipeline was designed to locate and recognise numerals in unconstrained environments. The system consists of four stages: Segmentation, Morphological repair, Candidate filtering, and Classification.

1. Segmentation: The input image was converted to grayscale and binarized using adaptive thresholding (Sensitivity 0.4).
2. Morphological Repair: Some of the handwritten digits had inconsistent strokes that did not connect fully. A morphological closing operation (imclose) was applied using a disk-shaped structuring element. This attempted to bridge the gaps between pen strokes, ensuring the digit was detected as a single connected component.
3. Candidate Filtering: A Connected Component Analysis (CCA) was performed using regionprops to extract bounding boxes. To reduce false positives, geometric filters were applied:
 - a. Solidity: Rejects solid blocks (like door handles) by requiring Solidity < 0.75.
 - b. Aspect Ratio: Rejects long, thin lines (like door gaps) by requiring a ratio < 4.
 - c. Area: Rejects background noise (dust, wood grain) smaller than 1000 pixels.
4. Classification: Valid regions were cropped, padded to preserve edge details, and resized to 28×28 pixels. These candidates were then classified using the CNN trained beforehand.



Results

The system's performance varied significantly depending on the background environments complexity.

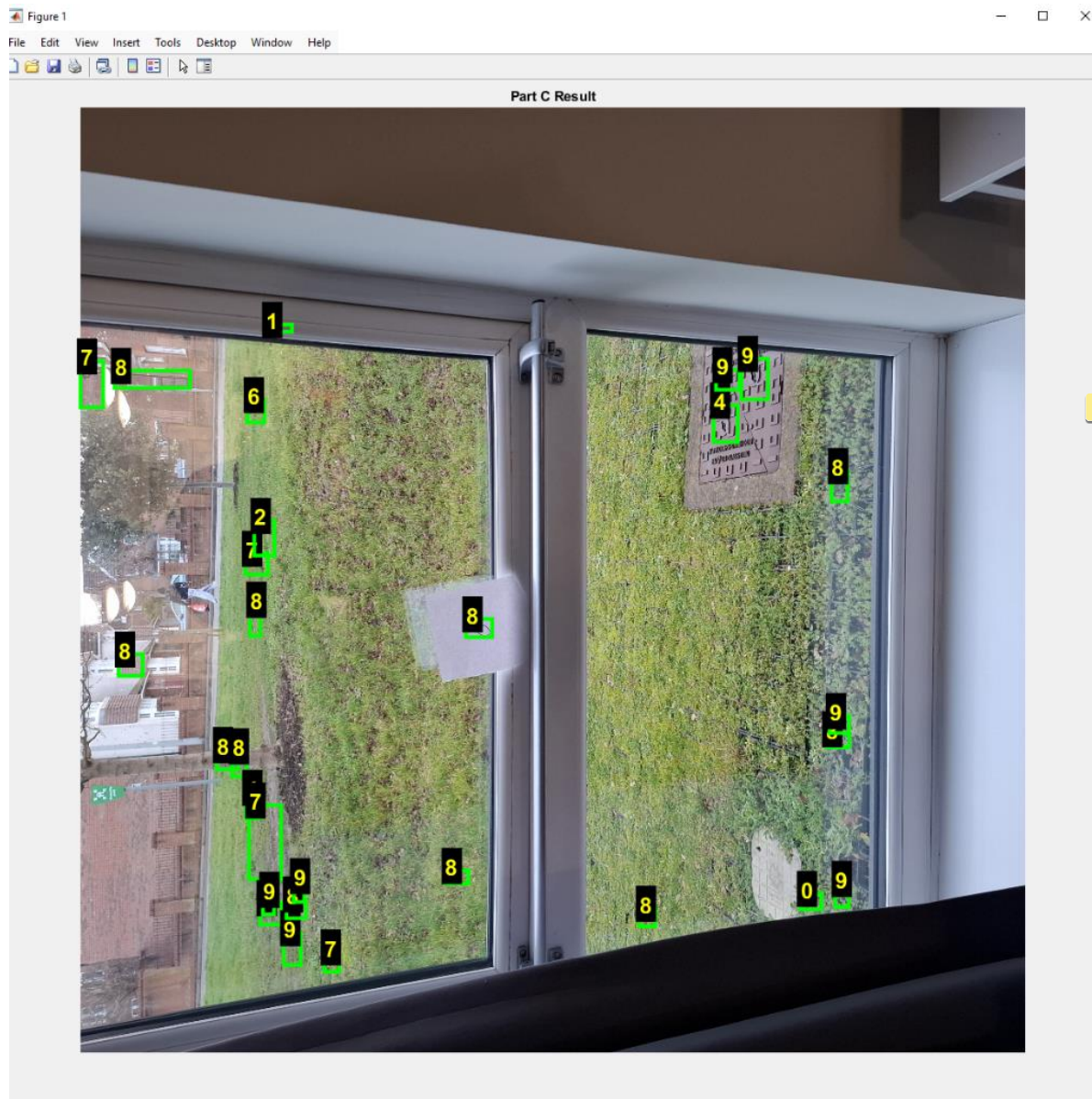


Figure 6

Success Case: As shown in Figure 6, the system successfully detected and correctly classified the digit '8' placed on the window. High contrast backlighting allowed the adaptive thresholding to isolate the digit effectively. However, the system also generated a lot of false positives in the background. The high-contrast texture of the grass and foliage was misinterpreted as "objects" by the blob analysis. Since these random shapes passed the area and solidity filters, the CNN was forced to classify them, resulting in random predictions.



Figure 7

Failure Case: In contrast, the system failed to correctly identify the digit on the door (Figure 7). Although the location of the paper was found, the classification was incorrect, which is probably due to the shape distortion of the digit. The rip in the paper and the angle of the handwriting likely altered the structural properties of the digit (e.g., changing its topology or aspect ratio). When this distorted shape was resized to the fixed 28×28 input of the CNN, it no longer resembled the standard training data, leading to a misclassification.

This comparison highlights a key limitation of the pipeline: while the detection logic is robust to lighting (finding the paper in both cases), the classification is brittle. It requires the detected object to be a clean, undisturbed shape. Background clutter (grass) and physical damage (ripped paper) significantly degrade the performance.



Extraction of 3-digit number

Method

The detection pipeline established in Part C was extended with a spatial sorting algorithm. Centroids of validated bounding boxes were sorted by x-coordinate to sequence digits Left-to-Right.

Results

The system was evaluated on a challenging sample containing handwritten digits on torn paper to test environmental robustness.

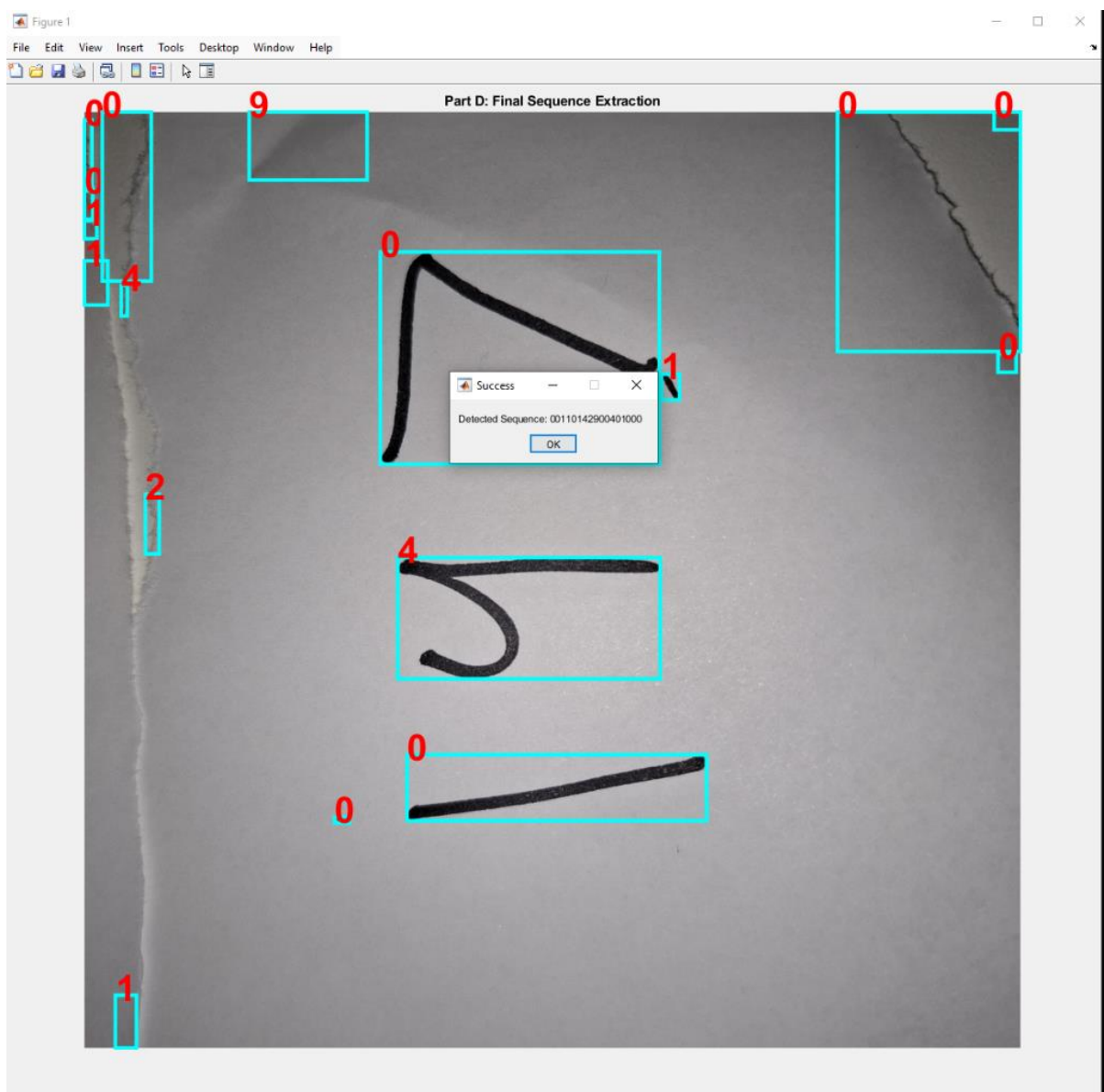


Figure 10

As shown in Figure 10, the system successfully localized the central digit '4'. However, it failed to distinguish between the handwritten ink and the vertical tears in the paper. The blob analysis algorithm interpreted the rips as valid objects because they shared geometric properties with digits (high contrast, vertical orientation, continuous lines).

Consequently, the CNN misclassified these tears as '1' or '0', and the sorting algorithm interleaved them into the final output string. This highlights a critical limitation: the system relies on geometric heuristics rather than semantic understanding. A robust solution for unconstrained environments would require a "background rejection" class to filter out structural noise before sorting occurs.



Completed system

Hardware Upgrades:

The current system runs on a laptop and uses a standard webcam, which is insufficient for a mobile robot operating day and night. The following specific hardware upgrades are proposed for the Kobuki base:

- Vision Sensor (Intel RealSense D435i): Features a Global Shutter to minimize motion blur and RGB-D depth sensing to differentiate flat targets from 3D obstacles. This allows the robot to tell the difference between a flat piece of paper (the number) and a 3D object (like the paper rips or door handles that caused errors in Part D).
- Computer (NVIDIA Jetson Orin Nano): The current MATLAB code is too slow for real-time collision avoidance. The Jetson is a small, powerful computer with a GPU that can run deep learning models on the robot itself without needing a laptop.
- Lighting (Wide-Angle White LED Array): To solve the need for night mode. The robot cannot rely on office lights being on. An automatic LED light will trigger when the camera detects darkness, ensuring the adaptive thresholding algorithm always has enough contrast to work.



Software & Algorithm Development:

To get from the current coursework code to a fully operational system, three key software changes are needed:

Better Detection (YOLOv8): Since Blob Analysis proved unreliable in clutter (Part D), a YOLOv8 object detector is proposed to learn contextual features of the target, reducing false positives.

Mapping (SLAM): The robot currently just "sees" a number. It needs to know where it is. I would integrate the vision system with the robot's LiDAR SLAM (Simultaneous

Localization and Mapping). When a number is recognized, the software will calculate its position in the room and save it to a digital map.

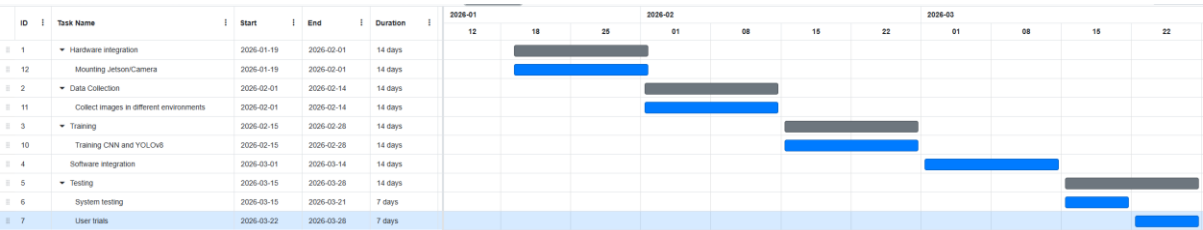
Tracking: To stop the robot from logging the same number 50 times as it drives past, a tracker (like DeepSORT) will be added to assign unique IDs to numbers.

Ethical Considerations:

Deployment raises GDPR concerns regarding employee surveillance and incidental recording of sensitive documents.

Mitigation: The system will be designed for "Privacy at the Edge." Video data will be processed instantly on the Jetson Nano and deleted immediately. Only the text result (e.g., "Number 724 found") will be saved. No video feed will ever be stored or sent to a cloud server.

Project management:



Conclusion

This project critically evaluated two distinct approaches to numeral recognition, demonstrating the superior robustness of deep learning over traditional hand-crafted features. While the detection pipeline successfully localized targets using adaptive thresholding, the classification stage revealed significant performance disparities.

Principal Findings:

- **Fragility of Global Descriptors:** Hu Moments proved ineffective for unconstrained tasks, achieving only ~30% accuracy on varied handwriting styles and angled shots due to a lack of 3D rotational invariance.
- **CNN Superiority:** The CNN achieved 63.2% accuracy on the mixed dataset, proving that learned hierarchical features generalize significantly better than static mathematical descriptors.
- **Semantic Gap:** The blob-analysis detection method struggled to distinguish between "ink" and "structural noise" (e.g., paper tears), leading to false positives in the output sequence.

Future Work: To improve system viability, future research should replace heuristic blob detection with a single-shot object detector (e.g., YOLOv8). This would introduce semantic understanding to the detection phase, preventing non-digit artifacts like tears or background textures from entering the classification pipeline.

Appendix

<https://github.com/mkwill12/AVLM>